D H A R

Volume II — Engineering Roadmap

STAGE-GAMMA

"Optimizing Dhar — earning the C-class performance commitment."

Document	Stage-gamma — Optimizing Dhar
Version	1.0
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Status	Ratified
Part of	The Canon of Dhar, Volume II
Depends on	Spec0, Spec1, Stage-beta (self-hosting proven)
Precedes	Stage-delta — Gearing Dhar

1. Mandate

Turn Stage-beta's correct-but-naive code generation into an optimizing compiler that can credibly stand behind the near-hardware, at-least-C-class performance commitment in Spec0 §2.2. Stage-gamma must not begin until Stage-beta is genuinely self-hosting (Stage-beta §4) — optimizing an unproven compiler produces unproven optimizations.

2. Scope

2.1 In scope

Category	Required passes
Representation	SSA construction
Register allocation	Graph-coloring allocation
Classical optimizations	Constant propagation, dead-code elimination, loop-invariant code motion, GVN/PRE
Interprocedural	Inlining, devirtualization of forge method calls (Spec1 §10.2)
Backend-facing	Vectorization, jump-table lowering for choose/is (Spec1 §8.3), architecture-specific tuning

2.2 Out of scope

- New language features or syntax changes — Spec1 is frozen for the duration of this stage.
- New backends beyond what is needed to demonstrate the optimizations; additional backends may proceed in parallel but are not this stage's mandate.

3. Required Architecture

Stage-gamma operates on Stage-beta's Universal IR, not on source text or the AST directly. This preserves the boundary Spec0 §4 establishes between architecture-agnostic and architecture-specific concerns: an optimization pass that needs to inspect source syntax rather than IR structure is almost certainly misplaced.

```
Stage-beta AST
     |
     v
Universal IR  ---->  [ Stage-gamma optimization passes operate here ]
     |
     v
Backend emission (unchanged interface from Stage-beta)
```

4. Acceptance Criteria

1. Correctness is non-negotiable: every test passing at the end of Stage-beta must still pass after every individual optimization pass is enabled, not only after the full pipeline runs together.

2. Each pass must be independently toggleable, so that a regression can be isolated to a single pass rather than the optimizer as a whole.

3. Each pass added must show a measured improvement on the language's own benchmark suite (Spec0 §7) relative to the baseline without it.

4. A numeric performance target relative to C is deliberately not fixed in this document: it should be set once Stage-beta's own unoptimized baseline is measured, and recorded here in a future revision rather than invented now without a baseline to invent it against.

5. Handoff to Stage-delta

Stage-delta may begin once Stage-gamma has landed SSA construction, register allocation, and dead-code elimination — the minimum viable optimizing pipeline — even if the remaining passes are still in progress. Stage-gamma and Stage-delta are not as strictly sequential as Stage-alpha, Stage-beta, and Stage-gamma are with each other.

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